

Writing Two-Column Proofs in Geometry

Geometry Worksheet · Grade 8–10

Name: _____

Date: _____

Learning Objectives

- Identify and apply properties of equality (addition, subtraction, multiplication, division) as reasons in a two-column proof
- Organize logical statements and corresponding justifications in a two-column proof format
- Apply reflexive, symmetric, transitive, substitution, and distributive properties to justify steps in algebraic and geometric proofs

Problems

1. Complete the two-column proof: Given $3x = 12$, state the reason that justifies the step $x = 4$.

$$3x = 12 \Rightarrow x = 4$$

2. Complete the two-column proof: Given $x + 9 = 15$, state the reason that justifies the step $x = 6$.

$$x + 9 = 15 \Rightarrow x = 6$$

3. Complete the two-column proof: Given $\frac{x}{4} = 5$, state the reason that justifies the step $x = 20$.

$$\frac{x}{4} = 5 \Rightarrow x = 20$$

4. Write a two-column proof for $5x + 12 = 47$ to show that $x = 7$, naming the reason for each step.

$$5x + 12 = 47$$

5. Complete the two-column proof: Given that segment $AB \cong$ segment AB , identify the property used.

$$AB \cong AB$$

6. Complete the two-column proof: Given that if $AB = CD$ and $CD = EF$, state the property that justifies $AB = EF$.



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$$AB = CD, CD = EF \Rightarrow AB = EF$$

7. Write a two-column proof for $2(x + 5) = 18$ to solve for x , naming the reason for each step.

$$2(x + 5) = 18$$

8. Write a two-column proof showing that if $m\angle A = m\angle B$ and $m\angle B = 45^\circ$, then $m\angle A = 45^\circ$, stating all reasons.

$$m\angle A = m\angle B, m\angle B = 45^\circ$$

9. Write a complete five-step two-column proof for $7x - 10(5 + 3x) = 20 - 2x$ to solve for x , naming every reason.

$$7x - 10(5 + 3x) = 20 - 2x$$

10. Write a two-column proof: Given that $\angle 1$ and $\angle 2$ are supplementary and $m\angle 1 = 3x + 10$, $m\angle 2 = x + 30$, prove that $x = 35$.

$$m\angle 1 + m\angle 2 = 180^\circ, m\angle 1 = 3x + 10, m\angle 2 = x + 30$$



Writing Two-Column Proofs in Geometry — Answer Key

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Answer Key

1. Answer: Division Property of Equality

- Statement 1: $3x = 12$ — Reason: Given
- Statement 2: $x = 4$ — Reason: Division Property of Equality (divide both sides by 3)

2. Answer: Subtraction Property of Equality

- Statement 1: $x + 9 = 15$ — Reason: Given
- Statement 2: $x = 6$ — Reason: Subtraction Property of Equality (subtract 9 from both sides)

3. Answer: Multiplication Property of Equality

- Statement 1: $\frac{x}{4} = 5$ — Reason: Given
- Statement 2: $x = 20$ — Reason: Multiplication Property of Equality (multiply both sides by 4)

4. Answer: $x = 7$; Reasons: Given, Subtraction Property of Equality, Division Property of Equality

- Statement 1: $5x + 12 = 47$ — Reason: Given
- Statement 2: $5x = 35$ — Reason: Subtraction Property of Equality (subtract 12 from both sides)
- Statement 3: $x = 7$ — Reason: Division Property of Equality (divide both sides by 5)

5. Answer: Reflexive Property of Congruence

- Statement 1: $AB \cong AB$ — Reason: Reflexive Property of Congruence
- Any segment (or angle) is congruent to itself by the Reflexive Property

6. Answer: Transitive Property of Equality

- Statement 1: $AB = CD$ — Reason: Given
- Statement 2: $CD = EF$ — Reason: Given
- Statement 3: $AB = EF$ — Reason: Transitive Property of Equality

7. Answer: $x = 4$; Reasons: Given, Distributive Property, Subtraction Property of Equality, Division Property of Equality

- Statement 1: $2(x + 5) = 18$ — Reason: Given
- Statement 2: $2x + 10 = 18$ — Reason: Distributive Property
- Statement 3: $2x = 8$ — Reason: Subtraction Property of Equality (subtract 10)
- Statement 4: $x = 4$ — Reason: Division Property of Equality (divide by 2)

8. Answer: $m\angle A = 45^\circ$; Reason: Substitution Property of Equality

- Statement 1: $m\angle A = m\angle B$ — Reason: Given
- Statement 2: $m\angle B = 45^\circ$ — Reason: Given
- Statement 3: $m\angle A = 45^\circ$ — Reason: Substitution Property of Equality (substitute $m\angle B = 45^\circ$ into Statement 1)



9. Answer: $x = -\frac{30}{21} = -\frac{10}{7}$; Reasons: Given, Distributive Property, Combine Like Terms, Addition Property of Equality, Division Property of Equality

- Statement 1: $7x - 10(5 + 3x) = 20 - 2x$ — Reason: Given
- Statement 2: $7x - 50 - 30x = 20 - 2x$ — Reason: Distributive Property
- Statement 3: $-23x - 50 = 20 - 2x$ — Reason: Combine Like Terms
- Statement 4: $-21x = 70$ — Reason: Addition Property of Equality (add $2x$ and 50 to both sides)
- Statement 5: $x = -\frac{10}{3}$ — Reason: Division Property of Equality (divide both sides by -21)

10. Answer: $x = 35$; Reasons: Definition of Supplementary Angles, Substitution, Combine Like Terms, Subtraction Property of Equality, Division Property of Equality

- Statement 1: $\angle 1$ and $\angle 2$ are supplementary; $m\angle 1 = 3x+10$, $m\angle 2 = x+30$ — Reason: Given
- Statement 2: $m\angle 1 + m\angle 2 = 180^\circ$ — Reason: Definition of Supplementary Angles
- Statement 3: $(3x + 10) + (x + 30) = 180^\circ$ — Reason: Substitution Property of Equality
- Statement 4: $4x + 40 = 180^\circ$ — Reason: Combine Like Terms
- Statement 5: $4x = 140$ — Reason: Subtraction Property of Equality (subtract 40)
- Statement 6: $x = 35$ — Reason: Division Property of Equality (divide by 4)

